$\Delta L = \alpha L_0 \Delta T$ (linear thermal expansion)

$\Delta V = \beta V_0 \Delta T$ (volume thermal expansion)

$Q = mc\Delta T$ (heat required to change temperature of a certain mass)

$Q = nC\Delta T$ (heat required to change temperature of a certain number of moles)

$Q = \pm mL$ (heat transfer in a phase change)

$H = \frac{dQ}{dt} = kA\frac{T_H - T_C}{L}$ (heat current in conduction)

$H = \frac{A(T_H - T_C)}{R}$ (heat current in conduction)

$R = \frac{L}{k}$ (relationship between thermal resistance and thermal conductivity)

$H = Ae\sigma T^4$ (heat current in radiation)

$H_{net} = Ae\sigma\left(T^4 - T_s^4\right)$ (net heat current in radiation)

$m_{total} = nM$ (total mass, number of moles, and molar mass)

$pV = nRT$ (ideal-gas equation)

$\left(p + \frac{a\,n^2}{V^2}\right)(V - n\,b) = n\,R\,T$ (van der Waals' equation)

$R = k\,N_A$ (gas constant, Boltzmann constant, Avogadro's number)

$M = N_A m$ (molar mass, Avogadro's number, and mass of a molecule)

$E_{kin} = \frac{m\,v^2}{2} = \frac{p^2}{2m}$ (kinetic energy)

$p = m\,v$ (momentum)

$E_{pot} = m\ g\ h$ (potential energy near the earth's surface)

$W = F\ s$ (work)

$K_{\text{tr}} = \frac{3}{2}nRT$ (average translational kinetic energy of an ideal gas)

$\frac{1}{2}m\left(\upsilon^2\right)_{\text{av}} = \frac{3}{2}kT$ (average translational kinetic energy of a gas molecule)

$\upsilon_{\text{rms}} = \sqrt{\left(\upsilon^2\right)_{\text{av}}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$ (root-mean-square speed of a gas molecule)

$\lambda = \upsilon t_{\text{mean}} = \frac{V}{4\pi\sqrt{2}r^2N}$ (mean free path of a gas molecule)

$C_V = \frac{3}{2}R$ (ideal gas of point particles)

$C_V = \frac{5}{2}R$ (diatomic gas, including rotation)

$C_V = 3R$ (ideal monatomic solid)

$f\left(\upsilon\right) = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2}\upsilon^2 e^{-m\upsilon^2/2kT}$ (Maxwell – Boltzmann distribution)

$W = \int_{V_1}^{V_2} p\,dV$ (work done in a volume change)

$\Delta U = Q - W$ (first law of thermodynamics)

$dU = dQ - dW$ (first law of thermodynamics, infinitesimal process)

$dU = n\ C_V\ dT$ (change of inner energy and temperature for ideal gas, all processes)

$C_p = C_V + R$ (molar heat capacities of an ideal gas)

$$\gamma = \frac{C_p}{C_V} \quad \text{(ratio of heat capacities)}$$

$p\,V^{\gamma} = constant$ (reversible, adiabatic process)

$T\,V^{\gamma-1} = constant$ (reversible, adiabatic process)

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right| \quad \text{(thermal efficiency of an engine)}$$

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|} \quad \text{(coefficient of performance of a refrigerator)}$$

$$e_{Carnot} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} \quad \text{(efficiency of a Carnot engine)}$$

$$K_{Carnot} = \frac{T_C}{T_H - T_C} \quad \text{(coefficient of performance of a Carnot refrigerator)}$$

$$\Delta S = \int_1^2 \frac{dQ}{T} \quad \text{(entropy change in a reversible process)}$$

$\Delta S = \frac{\Delta Q}{T}$ (entropy change in a reversible process at constant temperature)

$$S = k \ln w \quad \text{(microscopic expression for entropy)}$$